

Frontend UI / UX Specification

Audience: Frontend Engineers & UI/UX Designers

Goal: Design a professional, simple, intuitive UI that integrates all approved features across Panels 1–3, with minimal cognitive load and clear system feedback.

1. Design Principles (Non-Negotiable)

1. **Clarity over power** – users should never guess what to do next.
 2. **Progress visibility** – async AI systems must always show status.
 3. **One primary action per panel** – secondary actions stay subtle.
 4. **Deterministic feel** – system should feel predictable, not "chatty".
 5. **Fail visibly, recover cleanly** – errors should be actionable.
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2. Global Layout

Overall Structure

- **Three-panel layout (left → right)**
 - Panel 1: Documents
 - Panel 2: Prompt Chain
 - Panel 3: Run & Output

Global UI Elements

- Top bar:
 - App name
 - Session name / ID (read-only)
 - Lightweight user/profile menu
 - Toast system:
 - Success (green)
 - In-progress (neutral)
 - Error (red, dismissible)
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3. Panel 1 — Documents (Ingestion & Conversion)

Primary User Goal

Upload documents and ensure they are successfully converted.

UI Components

- **Input Type Selector**
 - Visible but disabled options ("Coming Soon"):
 - DOCX, TXT, MD, PNG

- Enabled: PDF only
- **Upload Button (Primary CTA)**
- Shows file count once selected
- **Document List**
- Columns:
 - Filename
 - Status badge (Queued / Processing / Converted / Error)
 - Action icon (delete if errored)
- Row is clickable → document detail drawer

UX Notes

- Conversion should auto-start after upload
 - Error rows must show short error reason
 - Deleting errored docs should feel safe and reversible (confirm modal)
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4. Panel 2 — Prompt Chain (Configuration)

Primary User Goal

Select or configure *what* logic will run on documents.

UI Components

- **Saved Chain Selector**
- Each chain card shows:
 - Chain name
 - Short description
 - Step count
- **Chain Detail View**
- Read-only by default
- Shows:
 - Ordered steps (R1...Rn)
 - Step descriptions

UX Notes

- No prompt editing inline (keeps UI clean)
 - Invalid or incomplete chains must be clearly labeled
 - Selecting a chain should feel lightweight (no modal overload)
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5. Panel 3 — Run & Output (Execution)

Primary User Goal

Run the chain and retrieve outputs confidently.

UI Components

- **Run Button (Primary CTA)**

- Disabled until:

- All docs converted
- Valid chain selected

- **Run Progress Table**

- Rows per document

- Columns:

- Filename
- Current step (R0...Rn)
- Status
- Token usage (in / out)

- **Completion Summary**

- Total docs processed

- Success vs failed count

- **Download Actions**

- Download per document
- Download all outputs

UX Notes

- Progress must update live (polling acceptable)
 - Token visibility builds trust and cost awareness
 - Failed documents should not block successful ones
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6. States & Feedback

Loading States

- Skeleton loaders preferred over spinners

Empty States

- Panel 1: "Upload documents to begin"
- Panel 2: "Select a saved chain"
- Panel 3: "Ready to run"

Error States

- Inline errors for doc-level issues
 - Toast for system-level issues
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7. Accessibility & Polish

- Keyboard navigable
 - Clear focus states
 - No dense text blocks
 - Consistent spacing & typography
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8. What NOT to Build (Important)

- No chat-style UI
 - No inline prompt editors
 - No hidden auto-runs
 - No multi-step wizards
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9. Frontend-Backend Contract Assumptions

- Frontend treats backend as source of truth
 - All async work is status-driven
 - Frontend never infers state
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10. Success Criteria

A first-time user should: 1. Upload PDFs without guidance 2. Understand conversion progress 3. Select a chain confidently 4. Run and download outputs without confusion

If any of these fail → UI is too complex.